Assignment 1: Exploratory Data Analysis on FIFA World Cup Squads

Course: EDA for AI · Session 1–2 · Total Marks: 40 (+ LeetCode practice)

Dataset: `fifa_world_cup_squads.csv`

Introduction

You are a Junior Data Analyst at GoalStat, a football analytics company. Ahead of the next tournament, the scouting team has handed you a raw export of 436 player records drawn from recent FIFA World Cups (2014, 2018 and 2022). They want you to explore player profiles, scoring and squad composition — and, just as importantly, to clean and prepare this messy export so it is ready for analysis and modelling.

Like most real-world data, this file is far from tidy: untidy column names, duplicate records, values stored as text that should be numbers, dates in several formats, missing values (some hidden in disguise), and even a few impossible values. Your job is to find and fix these issues using the techniques from Sessions 1 and 2.

Your goals are to:

- Explore and understand the dataset
- Inspect and select data using pandas (.loc / .iloc / Boolean masking)
- Identify and convert data types (numeric, object, datetime)
- Rename and restructure columns/rows
- Detect and remove duplicate records
- Identify and handle missing values (drop vs fill)
- Answer a few exploratory questions about the squads

Note: this dataset is synthetic and was created for training purposes. Player and country names are real, but all statistics are fictional.

Dataset Overview

The file contains the following columns (note that some column names are untidy on purpose):

Column	Description	Type
`Player_ID`	Identifier for each player record	integer
`Player Name`	Name of the player	text
`Country `	National team (note the trailing space in the name)	text
`Position`	Playing position (GK, DF, MF, FW)	text

`Club`	Club the player plays for	text
`Age`	Player age in years	numeric
`Matches_Played`	Matches played in that World Cup	numeric
`Goals`	Goals scored (stored as text — see Q5)	text
`Assists`	Assists made	numeric
`Minutes Played`	Minutes played in the tournament	numeric
`Market Value`	Estimated market value, e.g. “€84.0M” (text)	text
`Height`	Player height (mixed units — bonus)	text
`Is_Captain`	Whether the player is captain	text
`World_Cup_Year`	Tournament edition (2014 / 2018 / 2022)	integer
`Last Updated`	Date the record was last updated (text)	text

Section 1: Pandas Fundamentals & Exploration

Q1: Data Loading & Basic Inspection (5 marks)

1. Import the dataset into a pandas DataFrame.

2. Display each of the following:

- the first 10 rows of the data
- a structural summary of the dataset (use .info())
- a statistical summary of the numeric columns (use .describe())

3. Identify:

- the total number of rows and columns
- the data type (dtype) of every column

4. Separate the columns into two groups and print each list:

- numeric columns
- text (object) columns (hint: df.select_dtypes(include='number') / include='object')

Theoretical (3 marks)

1. Why is .describe() useful when exploring player data? Give one concrete example of something it reveals about this dataset (hint: look carefully at the min and max of Age).

2. The Goals column looks like it should be numbers, yet its dtype is 'object'. What does that tell you, and why is it a problem for analysis?

3. Why is it important to check data types before starting any analysis?

Q2: Data Selection & Subsetting — .loc[] and .iloc[] (5 marks)

1. Using .iloc[] (by position):

- select the first 10 rows
- select only the columns in the positions for Player Name, Country and Age

2. Using .loc[] (by label):

- display only the records from the 2022 World Cup (World_Cup_Year == 2022)
- for those rows, show only Player Name, Country and Age

3. Create a subset called regulars containing only players with Matches_Played of 6 or more. How many records are in it?

Q3: Filtering with Boolean Masking (3 marks)

1. Find all players older than 30 (Age > 30). How many are there?

2. Count how many players in the 2022 World Cup played more than 5 matches (combine two conditions — use & and wrap each condition in parentheses).

3. Count how many players made 3 or more Assists.

Theoretical (2 marks)

1. In your own words, what is the difference between .loc[] and .iloc[]?

2. Why is Boolean masking a useful skill when analysing player or squad data?

Section 2: Structural Cleaning & Missing Data (17 marks)

Q4: Missing Data Analysis (4 marks)

1. Identify the number of missing values in each column (use .isnull().sum()).

2. Report clearly which columns contain missing values and how many.

3. Some missing values are hidden in disguise — the Goals column uses 'N/A', '-' or '?', and Market Value uses 'Unknown' or '-'. Find these disguised values and replace them with a proper NaN (hint: df.replace([...], np.nan)). Then re-run .isnull().sum() and note how the counts change.

4. Theoretical: why is it important to check for missing values — including disguised ones — even when a dataset looks complete at first glance?

Q5: Identifying & Converting Data Types (4 marks)

1. Convert Goals to a numeric column (after handling the disguised missing values from Q4, use pd.to_numeric(..., errors='coerce')).

2. Convert Market Value (e.g. '€84.0M') into a number in millions — remove the '€' and 'M', treat 'Unknown'/'-' as missing, then convert to a float.

3. Convert Last Updated from text into a proper datetime (hint: pd.to_datetime(..., errors='coerce'). How many values could not be parsed and became NaT?

4. Age is currently a float because of its missing values. After you handle the missing values (Q8), convert Age to an integer type.

Theoretical: why does correct data typing matter? (Think about calculations, sorting, dates and feeding the data to a model.)

Bonus (optional): the Height column mixes units (e.g. '185', '1.85 m', '1,81', '190 cm', '?'). Write a small function to convert all of them to a single unit (centimetres).

Q6: Duplicate Records   (3 marks)

1. Check for fully duplicated rows using .duplicated() and count them.

2. Check whether any Player_ID is repeated (use subset=['Player_ID']). Should a Player_ID ever appear more than once? What would a repeated ID suggest?

3. Remove the exact duplicate rows, and report the row count before and after.

4. Theoretical: why are duplicate records dangerous in player / squad analysis?

Q7: Renaming & Restructuring Columns   (3 marks)

1. Several column names are untidy (extra spaces, mixed style). Using df.rename(...), give them clean names, for example:

- 'Country ' → 'country'   (note the trailing space in the original)
- 'Player Name' → 'player_name'
- 'Minutes Played' → 'minutes_played'   and   'Last Updated' → 'last_updated'

2. Remove a column you will not analyse (for example Minutes Played or Club) using df.drop(...). Confirm it is gone.

3. Restructure the table by setting Player_ID as the index, then reset it back to a normal column. (Reflect: is Player_ID a good index for this data? Why or why not?)

Q8: Missing Data Strategy & Impossible Values   (3 marks)

Decide how to handle the missing values, and justify every choice. Apply your strategy in code.

1. For the numeric columns (Age, Goals, Market Value, Assists, Matches_Played): fill the missing values with the mean or the median? Compare the two and justify your choice based on whether the column looks skewed.

2. For Club (a text column with missing values): fill with the mode, or use another approach? Justify.

3. Impossible values: use .describe() to spot values that cannot be real — for example an Age of 199 or 220, an Age of 0, or a negative Goals value. Replace these with NaN, then apply your fill strategy so they are corrected.

4. Theoretical: in general, when is dropping rows a better choice than filling them?

Section 3: Exploratory Insights (5 marks)

Using your cleaned dataset, answer the following. A short written insight matters as much as the code.

1. How many player records come from each World Cup year? (hint: value_counts() on World_Cup_Year)

2. After converting Goals to numbers, which player scored the most goals in the dataset? (hint: sort_values or idxmax)

3. Compare the average Goals scored in the 2022 World Cup versus the 2018 World Cup (use Boolean masking + .mean()).

4. After converting Market Value, compare the average market value of players older than 30 with those aged 30 and under (masking + .mean()). What does the difference suggest?

5. Observation & insight: run value_counts() on the country column. You will notice the same country appearing under different spellings and casing (e.g. 'Brazil', 'brazil', 'BRAZIL'). You do NOT need to fix this now — cleaning categorical text is the focus of the next session. In 2–3 sentences, describe the inconsistencies you see and one finding from your analysis above.

Bonus — preview of the next session (optional)

Use groupby to compute the average goals and average market value for each World Cup year.

```
df.groupby('world_cup_year')[['goals', 'market_value']].mean()
```

LeetCode Practice Problems

These reinforce the core Python (strings, lists, dictionaries and sets) from Session 1. Solve each as a function and test it on the examples.

1. Maximum Number of Balloons

Given a string text, count how many times you can spell the word “balloon” using the letters of text, where each letter can be used at most once. Return that maximum count.

```
text = "nlaebolko"         -> 1
text = "loonbalxballpoon"  -> 2
text = "leetcode"          -> 0
```

2. Longest Common Prefix

Given a list of strings, return the longest starting substring that all of them share. If there is no common prefix, return an empty string “”.

```
["flower","flow","flight"]  -> "fl"
["dog","racecar","car"]    -> ""
```

3. Contains Duplicate

Given a list of integers, return True if any value appears at least twice, and False if every value is unique. (Tip: a set links directly to today’s duplicate-detection topic.)

```
[1, 2, 3, 1]     -> True
[1, 2, 3, 4]     -> False
```

Submission Guidelines

- Use Google Colab for your solution.
- Use a separate, clearly labelled section for each question.
- Include markdown cells that describe your steps, assumptions and insights.
- Comment your code so each step explains what it does and why.
- Make sure every output is run and clearly visible before submitting.
- Submit your notebook as a .ipynb file (or share the Colab link with view access).

Tip: cleaning is iterative — inspect, fix one issue, then re-inspect. Aim for a final dataset with tidy column names, correct data types, no duplicates, and a deliberate, justified plan for every missing value.